

2.1 Variables and assignments ^

Settings

Beta features

Remembering a value

Here's a variation on a children's riddle.

PARTICIPATION ACTIVITY

2.1.1: People on bus.

For each step, keep track of the current number of people by typing in the num_people box (it's e

Start

You are driving a bus.
The bus starts with 5 people.

num_people:

1

2

3

4

5

Check

Next

The real riddle's ending question is actually, "What is the bus driver's name?" The subject usually says, "I started with YOU are driving a bus."

The box above serves the same purpose as a *variable* in a program, introduced below.

Variables and assignments

In a program, a **variable** is a named item, such as `x` or `num_people`, that holds a value.

An **assignment statement** assigns a variable with a value, such as $x = 5$. That statement means x is 5 during subsequent statements, until x is assigned again.

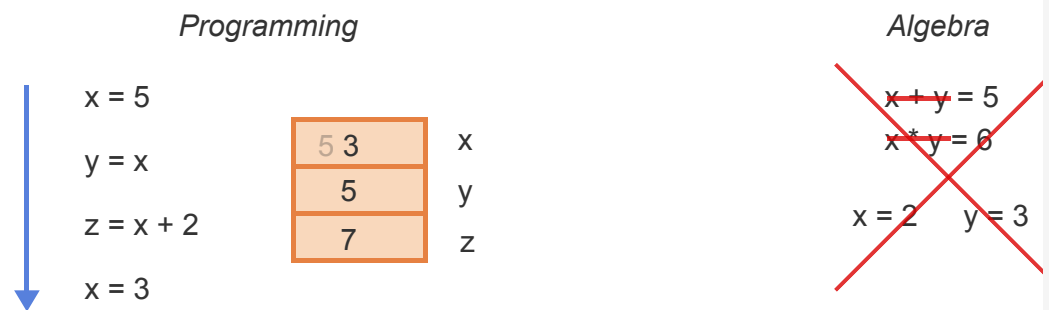
An assignment statement's left side must be a variable. The right side can be an expression, so a statement like $x = x + 2$ is valid. The 5, x , and $x + 2$ are each an expression that evaluates to a value.

PARTICIPATION ACTIVITY

2.1.2: Variables and assignments.

Start

☐ 2x speed



Captions ^

1. In programming, a variable is a place to hold a value. Here, variables x , y , and z are depicted.
2. An assignment statement assigns the left-side variable with the right-side expression's value.
3. $y = x$ assigns y with x 's value, which presently is 5. $z = x + 2$ assigns z with x 's present value.
4. A subsequent $x = 3$ statement assigns x with 3. x 's former value of 5 is overwritten and thus lost. y and z are unaffected, remaining as 5 and 7.
5. In algebra, an equation means "the item on the left always equals the item on the right." So we can determine $x = 2$ and $y = 3$.
6. Assignment statements look similar but have different meanings. The left side must be one variable.
7. The $=$ isn't "equals," but is an action that PUTS a value into the variable. Assignment statements are executed in sequence.

= is not equals

In programming, = is an assignment of a left-side variable with a right-side value. = is NOT equal is read as "x is assigned with 5" and not as "x equals 5." When a programmer sees $x = 5$, that goes into a box.

PARTICIPATION ACTIVITY

2.1.3: Valid assignment statements.

Indicate which assignment statements are valid.

1) $x = 1$

- ☐ Valid
☐ Invalid

2) $x = y$

- ☐ Valid
☐ Invalid

3) $x = y + 2$

- ☐ Valid
☐ Invalid

4) $x + 1 = 3$

- ☐ Valid
☐ Invalid

5) $x + y = y + x$

- ☐ Valid
☐ Invalid

PARTICIPATION ACTIVITY

2.1.4: Variables and assignment statements.

Given variables x , y , and z :

1) $x = 9$

$y = x + 1$

What is y ?

Check

Show answer

2) $x = 9$

$y = x + 1$

What is x ?

Check

Show answer

3) $x = 9$

$y = x + 1$

$x = 5$

What is y ?

Check

Show answer

CHALLENGE ACTIVITY

2.1.1: Enter the output of the variable assignments.

Note: Each `print()` includes a newline, so create a newline after your output by pressing *Enter* or *Return*.

763650.1443318.qx3zqy7

Start

Type the program's output

```
x = 7
y = 5
print(x, y)
```

7 5

1	2	3	4	5	6
---	---	---	---	---	---

Check

Next

View solution ▼ (Instructors only)

Assignments with variable on left and right

Because in programming = means assignment, a variable may appear on both the left and right as assigned with $6 + 1$, or 7. The statement overwrites the original 6 in x.

Increasing a variable's value by 1, as in $x = x + 1$, is common and known as **incrementing** the variable.

PARTICIPATION ACTIVITY

2.1.5: A variable may appear on the left side and right side of an assignment statement.

Start

☐

2x speed

```
x = 1
```

```
x = x * 20
```

```
x = x * 20
```

```
print("A person with measles may cause", x)
```

```
print("people to be infected in two weeks.")
```

1 20 400 x

A person with measles may cause 400 people to be infected in two weeks.

Captions ^

1. A variable may appear on both sides of an assignment statement. After $x = 1$, then $x = x * 2$ overwriting x 's previous 1.
2. Another $x = x * 20$ assigns x with $20 * 20$ or 400, which overwrites x 's previous 20.
3. Only the latest value is held in x .

**PARTICIPATION
ACTIVITY**

2.1.6: Variable on both sides.

Indicate the value of x after the statements execute.

- 1) $x = 5$
 $x = x + 7$

Check[Show answer](#)

- 2) $x = 2$
 $y = 3$

$x = x * y$
 $x = x * y$

Check[Show answer](#)

- 3) $y = 30$
 $x = y + 2$
 $x = x + 1$

Check[Show answer](#)

4) Complete this statement to increment y by 1:

y = ____

Check

Show answer

**CHALLENGE
ACTIVITY**

2.1.2: Trace the variable value.

763650.1443318.qx3zqy7

Start

Select the correct values for x, y, and z after the following assignments execute.

x = 6

y = 4

z = 5

x = 3

What is x?

Pick ▼

What is y?

Pick ▼

What is z?

Pick ▼

1

2

3

4

Check

Next

View solution ▼ (Instructors only)

**CHALLENGE
ACTIVITY**

2.1.3: Variables and assignments.

763650.1443318.qx3zqy7

Start

Assign variable num_clementines with 40.

The output is: 40

[Learn how our autograder works](#)

```
1  
2 """ Your code goes here """  
3  
4 print(num_clementines)
```

1

2

3

Check

Next level

View solution ▼ *(Instructors only)*